Yidi Zhao

Ph.D. Candidate | Department of Economics | Penn State University

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Research Interests

Econometrics: Model Misspecification, Measurement Error, Statistical Decision Theory

Applied Econometrics: IO, Health, Labor

Education

Ph.D. in Economics, Penn State University — *May 2027* (*expected*)
Department of Economics

M.A. in Economics, Peking University — *Jul. 2020*
National School of Development

B.A. in Economics, Central University of Finance and Economics — *Jul. 2017*
School of Finance

Research

Job Market Paper

Robust Inference under Local Distribution Shift in Wasserstein Distance

Abstract: This paper studies the robustness of estimators to small departures of the true data-generating process from a reference model, which defines the parameter of interest. We measure distribution shift using the quadratic Wasserstein distance, which captures movements of the observations in the sample space due to various factors like measurement error. We consider local deviations within shrinking Wasserstein neighborhoods so that both sampling uncertainty and distributional bias matter asymptotically. Within this framework, we first derive local asymptotic minimax bounds for the regular estimator classes. We then construct attaining procedures, including a one-step estimator for regular parametric models and a new hybrid empirical-likelihood–optimal-transport (EL–OT) estimator for models defined by unconditional moment restrictions. We illustrate our method in the highway-and-trade setting of Duranton et al. (2014) with instrumental-variable moment restrictions and find that point estimates remain stable while bias-aware confidence intervals widen as larger Wasserstein departures are allowed.

Work in Progress

Multinomial Logistic Regression with Sample Selection

Joint work with Onil Boussim (*Cal Poly SLO*)

Nonparametric Tests for Proprietary Information at First-Price, Common Value Auctions

Teaching

Instructor, Department of Economics, Penn State University

ECON 106: Statistical Foundations for Econometrics (UG) *Summer 2026*

Teaching Assistant, Department of Economics, Penn State University *Fall 2021 – Spring 2025*

ECON 454: Economics of Mergers (UG); ECON 402: Decision Making and Strategy in Economics (UG); MA ECON 510: Econometrics (Grad)

Teaching Assistant, National School of Development, Peking University

Advanced Microeconomics II (Graduate core course) *Spring 2019*

Social Science Survey Data Analysis (UG) *Fall 2018*

Skills

Programming: Stata, Python, and Julia.

Language: English (fluent), Chinese (native).

References

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